

WASP-104b

R-2628

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_150513 · created 2026-08-29T00:03:41+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457155.66909 +/- 0.00047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.034
Transit depth (Rp/Rs)^2	2.21 +/- 1.0 [%]
Orbital Inclination (inc)	85.18 +/- 1.77
Airmass coefficient 1 (a1)	0.955 +/- 0.02
Airmass coefficient 2 (a2)	0.032 +/- 0.027
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	#3 - [494, 51]
Optimal Method	PSF photometry
Transit Duration (day)	0.0564 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.149 ± 0.034 vs 0.1214 (deviation 0.68σ)
Duration fit vs published	0.0564 d vs 0.0742 d (deviation 2.48σ)
Mid-time O–C	-1.4 min
Depth SNR	3.7
Residual scatter	2.09 %

Light curve

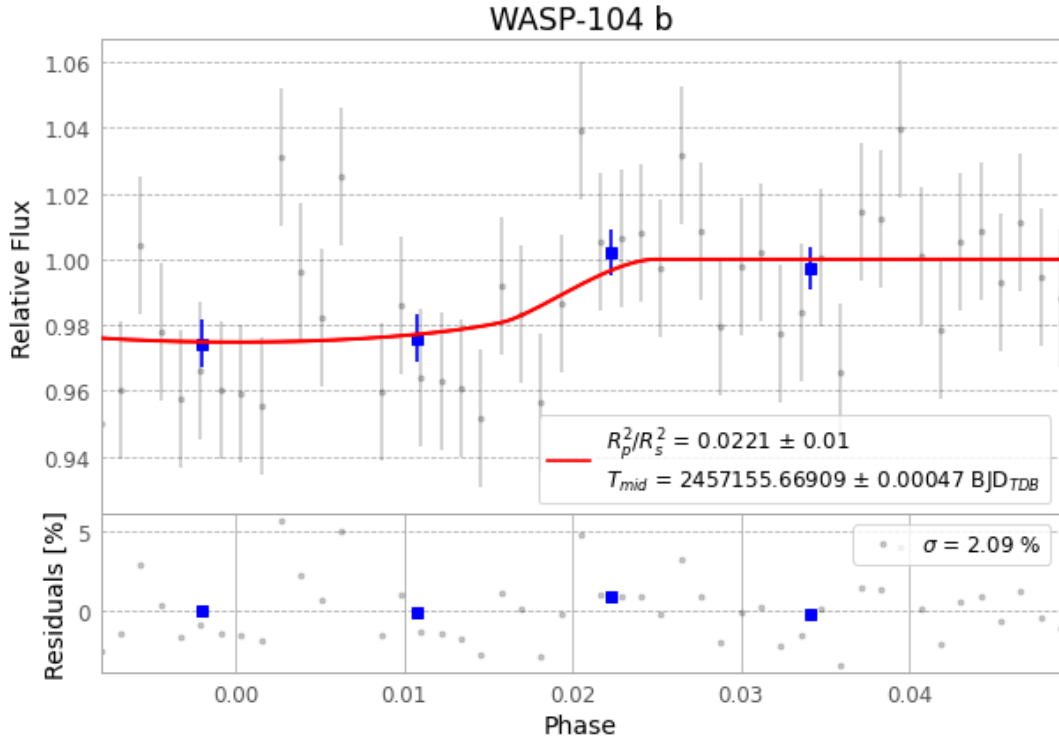


Figure 1 — FinalLightCurve_WASP-104 b_2015-05-12.png (SHA-256 20c4a627623904b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 198], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6050b338e9856c4d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

52 FITS frames (34 MB), WASP-104150513033012.FITS → WASP-104150513060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-150513.log (69db0d25c9f5...), FinalLightCurve_WASP-104 b_2015-05-12.png (20c4a6276239...), FinalLightCurve_WASP-104 b_2015-05-12.csv (575e8ffcc8c8...)

Compute resources

Wall time 133.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:03Z.